Data Analyst (R) Technical Assessment: Brief Report

Dataset: Nurse Stress Dataset (Empatica E4), Dryad, doi:10.5061/dryad.5hqbzkh6f. This report follows the brief's parts (1 to 7, then final deliverables); assumptions are stated where they are used.

> **In one minute. Question:** can a wrist-worn sensor tell when a hospital nurse is stressed? **Answer:** not reliably. After allowing for which nurse wore the band, skin conductance, heart rate, skin temperature and movement did not separate "high stress" periods from the rest, and who the nurse is explained more than any signal. **Key numbers:** 15 nurses, 609 recordings (about 1,251 hours); 214 survey periods modelled (161 high stress, 53 not); strongest link between any signal and reported stress 0.13 (0 = none, 1 = perfect); signals explain about 5% of the picture, which nurse it is about 17%.

Part 1: Data Ingestion & Integration

- **Multiple file types:** 609 zipped per-session CSV recordings from 15 nurses (skin conductance EDA, heart rate HR, skin temperature TEMP and movement ACC; line 1 is the start time, line 2 the sample rate) plus one Excel survey sheet (358 rows: nurse ID, start/end time, date, stress level 0 to 2). BVP, IBI and the empty tags file were not needed.
- **Merge keys:** signals are joined on the timestamp after averaging each to one reading per second (heart rate is recorded once a second, so this is the common speed); each survey is joined by nurse ID plus start and end time.
- **Assumptions:** (1) survey times are local Chicago time and were converted to the sensors' UTC clock; testing hour shifts per nurse showed no shift is best for all 15 nurses afterwards, lifting matched labels from 2% to 7.9% of seconds. (2) Movement in g = raw value / 64 (about 1 g when still). (3) If two survey rows overlap, the later row in the sheet wins (all overlapping pairs carry the same level).
- **Result:** all 609 recordings loaded; 1,501 repeated seconds removed; 4,505,058 rows (one per nurse per second); 7.9% of seconds carry a stress label. Missing values and inconsistencies are covered in Part 6. **Deliverables:** `data_loading.R`.

Part 2: Data Understanding & Feature Engineering

Key variables: skin conductance (EDA, uS; rises with arousal), heart rate (bpm; also rises with activity), skin temperature (degrees C; also shows the band is worn), movement (g; separates exercise from stress). **Derived features** (the brief asks for 3; seven built):

Feature	Plain meaning	Type
`HR_roll60`, `acc_roll60`	Heart rate and movement averaged over the last minute	Rolling average
`EDA_roll30`	Skin conductance averaged over the last 30 seconds (its recent baseline)	Rolling average
`HR_sd60`	How much heart rate fluctuates over the last minute (not clinical heart rate variability)	Stress proxy
`EDA_phasic`	Skin conductance minus its recent baseline (a quick arousal spike)	Stress proxy
`EDA_slope`	How fast skin conductance is changing (left out of the model: moves with `EDA_phasic`, correlation 0.91)	Stress proxy
`day_part`	Morning, afternoon or evening/night (local time); built but not used in the model	Time segment

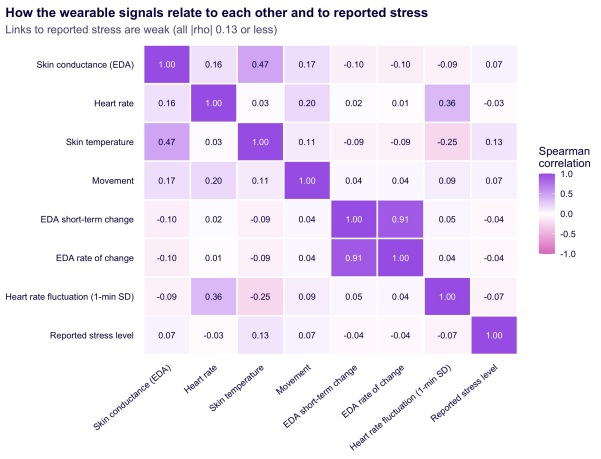
Averages look back only within one recording. Signals were also averaged into one-minute chunks (75,687 chunks, 6,152 with a stress label). **Deliverables:** `features.R`.

Part 3: Correlation Matrix Heatmap

Method: Spearman correlation (compares rankings), chosen because skin conductance and movement are skewed with extreme values, body-signal links need not be straight lines, and stress level is a 0 to 2 score. **Missing values:** each pair uses every one-minute chunk where both values exist (pairwise deletion). Drawn with ggplot2.

- Skin temperature and skin conductance move together (0.47), mostly because both reflect the band being worn.
- The two skin conductance change measures are nearly the same thing (0.91).
- Heart rate rises with movement (0.20), as expected.
- Every link to reported stress is weak: 0.13 at most (skin temperature), 0.07 or less for the rest.

Deliverables: heatmap (`outputs/heatmap.png`, `heatmap_branded.png`).

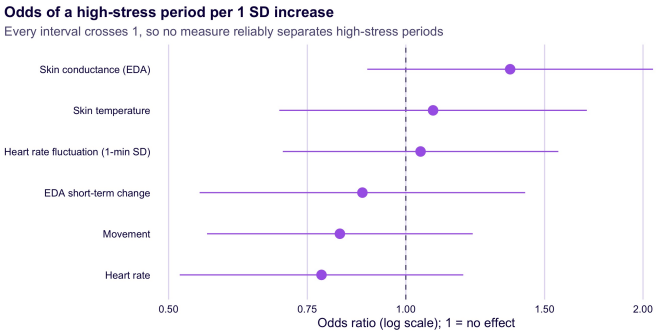


Part 4: Statistical Analysis

Technique: mixed-effects logistic regression (`stats.R`). **Hypothesis:** *no effect (H0):* average signals during a survey period are unrelated to whether it was rated high stress, once the nurse is accounted for; *effect (H1):* higher arousal signals make a high-stress period more likely (0.05 level). **Why this method:** the outcome is yes/no, and each nurse appears in many survey periods, so a model giving each nurse their own baseline is needed. One row per survey period, because the stress label is constant within a period.

Signal (per 1 SD higher)	Odds ratio	95% range	p
Skin conductance (EDA)	1.36	0.89 to 2.06	0.152
Heart rate	0.78	0.52 to 1.18	0.244
Skin temperature	1.08	0.69 to 1.70	0.728
Movement	0.82	0.56 to 1.22	0.331
EDA short-term change	0.88	0.55 to 1.42	0.601
Heart rate fluctuation	1.04	0.70 to 1.56	0.834

Odds ratio above 1: a higher reading made high stress more likely; 1 = no difference. Output also saved in `outputs/model_odds_ratios.csv`.



Assumptions validated: signals not too similar (highest variance inflation 1.45, limit 5: passed); model fitted properly (converged, no warnings: passed); repeated measures handled via the nurse baseline (about 17% of the variation); calibration imperfect (80% of residuals inside bounds, about 95% ideal); 13 influential points flagged, but the test is not valid for this model type, so no action; sample borderline (53 "not high" periods for six signals, about 8.8 each, guide 10). **Interpretation:** none of the signals reliably tells high-stress periods from the rest (every range includes 1, so "no effect" is not rejected); skin conductance came closest (about 1.4 times the odds, wide range); which nurse wore the band explained more than any signal. **Deliverables:** model output.

Part 5: Insight Generation

1. **A wristband alone is a weak stress marker:** links are 0.13 at most and no signal stands out, so a simple threshold would not reliably find stressful periods.
2. **The nurse matters more than the signals:** about 17% of the difference between high and lower stress periods is between nurses, against about 5% for all signals together, which favours personal baselines.
3. **Reports are skewed and sparse:** 73% of usable surveys (177 of 242) were the top stress level and only 7.9% of recorded time is labelled, leaving few calm periods to compare (53 of 214).

Part 6: Data Quality Check (Mandatory Gate)

`data_quality.R` runs second in the pipeline, straight after loading and before features, correlations or models.

Data Quality Considerations

Issue	Why it matters	What was found and done
Survey and sensor clocks differ	Wrong alignment hides real signal	Only 2% of seconds matched at first; converting local time to UTC lifted this to 7.9%.
Different sampling speeds	Signals cannot be compared second by second	Averaged to one reading per second.
Impossible values	Off-wrist readings look like extreme physiology	Blanked: skin temperature outside 25 to 42 C (22,202 s, all too cold), skin conductance outside 0.01 to 60 uS (128,973 s, 2.9%, all too low), heart rate outside 30 to 220 (none).
Repeated seconds	Overlaps would double-count	1,501 removed; none remain.
Missing values, gaps, units	Gaps spoil rolling averages; raw movement counts mean nothing	Under 0.15% missing per signal; no gap over 5 seconds within a recording; movement divided by 64 to give g.
Surveys with no sensor data	Nothing to analyse	28 of 242 distinct periods have none (clustered on certain dates, pointing to missing recordings); left out.
Duplicate or blank survey rows	Could double-count or leave no outcome	3 exact duplicates (same levels, no label changed); 113 of 358 rows have "na" for stress level and were dropped (cause not stated).

Survey rows from file to model: 358, minus 113 with no level = 245; minus 3 duplicates = 242; minus 28 with no sensor data = **214 modelled periods**.

Part 7: Findings Presentation

`docs/Sanius_Nurse_Stress_Findings.pptx`: slides for a non-technical healthcare audience, in Sanius colours and logo style taken from the brief. **Analysis:** slide 1 (question, 15 nurses, 214 periods, four signals, how the test works) and slide 2 (charts). **Findings:** slide 2 (weak signal link, nurses differ, few calm labels). **Interpretation, limitations, next steps:** slide 3; slide 4 compares the results with the dataset paper.

Limitation	Next step
Only 214 periods (53 not high stress); one reported level per period; the cutoff at level 2 is an analyst choice	Collect more, including calm periods; test a cutoff at level 1
Period averages hide change within a period; no personal baselines; stressor types not used	Model within-period change; compare each nurse with their own normal; test stressor types
Associations only, not cause and effect	Validate on new nurses before any real use

Final Deliverables

- **R scripts:** `config.R`, `data_loading.R`, `data_quality.R`, `features.R`, `correlation.R`, `stats.R`, `plots_branded.R`, `run_all.R`
- **Report:** `Report.pdf` (brief part-wise analysis)
- **Visualisations:** `outputs/heatmap.png`, `forest_plot.png` and `_branded` versions
- **README:** `README_documentation.md` (detailed documentation of analysis, assumptions and results)
- **Presentation:** `docs/Sanius_Nurse_Stress_Findings.pptx`
- **To reproduce:** put the dataset in `data/`, open the project in R and run `source("run_all.R")` (about 3 minutes)